

The empirical recurrence for OEIS A300962, recovered from its tail

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Abstract

OEIS A300962 records “Empirical recurrence of order 75” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 107$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 4$, so the recurrence is proved for every $n \geq 79$. Its last coefficient is $c_{75} = 4 \neq 0$, so no further tail satisfies anything shorter; any order-75 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A300962 is “Number of $n \times 4$ 0..1 arrays with every element equal to 0, 2, 3, 4 or 6 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

1, 16, 44, 327, 1653, 10135, 60730, 366981, ...

The entry states:

Empirical recurrence of order 75 (see link above).

As of the “Last modified” line on the live entry (Oct 29 2023 02:22:05, revision 8) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 107$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 107$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 75: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 4$ is the first whose minimal order is exactly the stated 75.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 214$ exact terms from $n = 4$ on, it returns order 75 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{75} c_i a(n-i),$$

c_1	7	c_{20}	2207229	c_{39}	888128949	c_{58}	12058929
c_2	7	c_{21}	3269336	c_{40}	471576026	c_{59}	2639039
c_3	−78	c_{22}	−4749356	c_{41}	97709105	c_{60}	−1234208
c_4	−18	c_{23}	−12509855	c_{42}	−132638759	c_{61}	−915208
c_5	67	c_{24}	−22963047	c_{43}	−42314099	c_{62}	−1142902
c_6	438	c_{25}	−48874112	c_{44}	−41312849	c_{63}	128801
c_7	925	c_{26}	−40924546	c_{45}	−8008538	c_{64}	−32693
c_8	−3338	c_{27}	6828058	c_{46}	−194273383	c_{65}	126649
c_9	6362	c_{28}	105635640	c_{47}	−216716523	c_{66}	24378
c_{10}	6569	c_{29}	226325077	c_{48}	−147687478	c_{67}	20546
c_{11}	−35360	c_{30}	308474546	c_{49}	21959105	c_{68}	−1928
c_{12}	−1277	c_{31}	221470476	c_{50}	190228884	c_{69}	−4655
c_{13}	−13727	c_{32}	−58059356	c_{51}	146393108	c_{70}	−2255
c_{14}	−82094	c_{33}	−459175485	c_{52}	−6259264	c_{71}	−225
c_{15}	−41665	c_{34}	−803513349	c_{53}	−32316511	c_{72}	311
c_{16}	132998	c_{35}	−743102636	c_{54}	−70259381	c_{73}	104
c_{17}	542019	c_{36}	−359851659	c_{55}	−30249846	c_{74}	24
c_{18}	822719	c_{37}	290702199	c_{56}	4129806	c_{75}	4
c_{19}	1230111	c_{38}	717705616	c_{57}	11126545		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 79$, and it is the recurrence OEIS A300962 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 4$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{75} - c_1 x^{75-1} - \dots - c_{75}$. Its constant term is $-c_{75} = -4$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of

terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 75 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 75, so $\deg q = 75$, and it is monic. It holds from some index t on. From $\max(t, 4)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 75, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 21 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 75, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 4.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-75 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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